DER Analytics - Requirements and Architecture

DER Analytics - Site-Level Analytics Engine and Application

Software Requirements Specification | System Architecture | Clinical and Aggregate Reporting Analytics | UI/UX | Acceptance and Evaluation
Version: 1.0 (implementation baseline)
Target: DER Server and DER Client; interoperable with DHIS2 2.41, 2.42, 2.43 and 2.44
Status: Design specification; not an assertion of native DHIS2 execution parity or a completed implementation
Date: 10 November 2026

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1. Executive summary and design decisions

DER Analytics is a **site-local, offline-capable analytics engine and web application** for event programs, tracker enrollments, tracked entities, program indicators, program-stage data, aggregate datasets and departmental operations. Its principal job is to provide reliable, explainable, near-real-time results from the **site's committed data**, even while the central DHIS2 instance or the WAN is unavailable. It provides authorised drilldowns back to the records that contributed to each number. It must **not** pretend that a local partial dataset is the definitive national DHIS2 analytics output.

Decisions:

1. The **DER Server** owns canonical site analytics state, an incremental computation pipeline, snapshots, query authorisation, data lineage, and the publication bridge to aggregate reporting. DER Client provides views and offline cached authorised analyses; a browser is not the authority for computing facility-wide statistics.

2. Use a **DHIS2-compatible metadata-aware expression compiler**, not JavaScript `eval` or direct arbitrary SQL. A support matrix explicitly lists compatible expressions, missing functions and unsupported syntax per DHIS2 version. Unsupported indicators are flagged, not guessed.
3. Distinguish **live provisional**, **locally verified**, **published to aggregate workspace**, **approved for submission**, and **reconciled with central DHIS2** values.
4. Partition calculations by site, source organisation unit, event/enrollment date interpretation, source programme, clinical dimensions and configurable period. Respect source ownership, write permissions, and metadata snapshots.
5. Never confuse **visits, events, enrollments, unique patients, admissions, episodes, discharges, deaths, diagnoses, and cases**. Every metric defines grain, counting unit, denominators, event status, date attribution, org unit and deduplication policy.
6. Adopt **incremental upsert/retraction**: when a patient event changes, the previous contribution is removed before the new version is applied. Replay, deletion, transfer and conflicts must not double count.
7. A drilldown uses an explainable **lineage query** from a computed cell to permitted source contributions. Patient-level details require strict role-based authorisation and audit trails.
8. Create aggregate reporting proposals only via explicit **approved indicator-to-data-element mappings**, never by assuming that every tracker program indicator is equivalent to a DHIS2 aggregate data element.

Scope note: This specification describes intended DER features. It does not change the official DHIS2 APIs or assert full parity with every expression, period type or reporting rule. Validate each integration against the actual target server.

2. Relationship to the existing DER hospital design

DER Capture manages programmes, tracker enrollments and events. DER Distributed Aggregate Capture manages a parent dataset, native DHIS2 sections, logical subsections, service-point/device assignments, validation and approval. DER Analytics consumes committed versions from both and computes local measures and authorised reportable aggregates. Source data are never replaced by analytics output.

```
flowchart TD
 A[DER Capture: Event and Tracker] --> J[Durable Change Journal]
 B[DER Aggregate Capture: 15 parent datasets] --> J
 C[Metadata and DHIS2 Sync] --> M[Versioned Metadata Catalog]
 J --> I[Idempotent Analytics Ingestion]
 M --> X[Validated Definitions and Compiler]
 I --> F[Canonical Facts and Contributions]
 X --> E[Incremental Tally and Indicator Engine]
 F --> E
 E --> S[Period and Dimension Aggregates]
 S --> Q[Secured Query and Drilldown API]
 Q --> U[DER Analytics Web App]
 S --> P[Aggregate Mapping and Review Proposal]
 P --> B[DER Aggregate Review and Approval]
 B --> D[DHIS2 Data Value Sets Sync]
```

2.1 Existing parent aggregate reporting catalogue

The previously prepared general-hospital package contains 15 parent datasets, 175 native sections, and 1,039 aggregate data elements. The table below is extracted from that package and **does not constitute a claim that each listed measure is automatically derivable from Tracker**.

Code	Parent dataset	Period	Sections	Distinct section DEs
OPDS0001	Outpatient Department - Monthly Report	Monthly	39	238
IPDS0001	Inpatient Department - Monthly Report	Monthly	25	188
ERDS0001	Emergency Department - Monthly Report	Monthly	11	87
MTDS0001	Maternity and Newborn - Monthly Report	Monthly	10	58
ANDS0001	Maternal, Child and Reproductive Health - Monthly Report	Monthly	14	54
IMDS0001	Immunisation - Monthly Programme Report	Monthly	3	12
HVDS0001	HIV Care - Monthly Programme Report	Monthly	5	23
TBDS0001	Tuberculosis and Leprosy - Monthly Programme Report	Monthly	5	25
LBDS0001	Diagnostic Services - Monthly Report	Monthly	15	86

Code	Parent dataset	Period	Sections	Distinct section DEs
SUDS0001	Surgery, Procedures and Sterilisation - Monthly Report	Monthly	10	49
PHDS0001	Pharmacy and Supply Chain - Monthly Report	Monthly	10	50
BLDS0001	Billing and Financial Services - Monthly Report	Monthly	5	23
ICDS0001	Infection Prevention and Patient Safety - Monthly Report	Monthly	5	21
HMDS0001	Hospital Management, Records and Quality - Monthly Report	Monthly	15	53
SRDS0001	Communicable Disease Surveillance - Weekly Report	Weekly	3	72

These parent datasets can contain many logically assigned sub-workflows. DER Analytics must expose drilldown by parent dataset -> service group -> native section -> DER subsection -> metric/disaggregation -> authorised contributing source records.

2.2 Systems of record and authority

Concern	System of record	DER Analytics responsibility
Patient, enrollment, event	DER site operational store while pending; central DHIS2 after accepted reconciliation per agreed rules	Read-versioned projections, source links
Hospital stock deduction	DER Inventory or designated eLMIS ledger	Read-only movement and balance-derived metrics
DHIS2 metadata	Central DHIS2 metadata, held as versioned local snapshots	Interpret semantics using exactly the compiled snapshot
Department report section	DER Aggregate Capture workspace	Suggest computable numbers, show provenance and workflow status
Submitted aggregate value	DHIS2 dataValueSet once acknowledged	Reconcile and show differences; never silently overwrite
Indicator definitions	DHIS2 programme indicators / governed DER derived indicators	Compile, execute or declare unsupported

3. Goals, users, boundaries, assumptions

Goals: Near-real-time local summaries; clinical/operational monitoring; weekly/monthly/other supported periods; longitudinal trend and cohort analytics; diagnosis/case and mortality reporting; explainable drilldowns; offline site continuity; workload visibility; explicit aggregation publication; verifiable consistency with central results.

Personas: HMIS officer; hospital administrator; programme manager; department head; clinician with a treatment relationship; pharmacist; laboratory lead; site analyst; quality officer; system administrator; privacy auditor. Each persona sees only authorised organisations, programmes, fields and record levels.

Not in scope for the initial engine: population-level national deduplication across disconnected sites; clinical decision-making without review; automatic medical diagnosis; arbitrary execution of unknown DHIS2 functions; public disclosure of patient identities; a replacement for central DHIS2 analytics infrastructure; unrestricted query access to personally identifiable health data.

Assumptions: The site can access a durable local database and metadata snapshots; recording events and enrollments preserves stable IDs and change versions; facility time zone is configured, usually Africa/Kampala for Uganda deployments; explicit retention and lawful-use policies exist.

4. Detailed functional requirements

The IDs below are stable software requirements, not DHIS2 metadata codes. `MUST` means required before production; `SHOULD` means recommended for a full release; `MAY` means optional extension.

4.1 Metadata discovery, compatibility and expression engine

ID	Level	Requirement
`AN-FR-001`	MUST	Ingest DHIS2 programme, program stage, data element, option, tracked entity attribute, indicator, category, org unit and reporting metadata by immutable revision.
`AN-FR-002`	MUST	Parse the configured program indicator analytics type (EVENT/ENROLLMENT), aggregation type, expression, filter, program, period boundaries and organisation-unit attribution.

ID	Level	Requirement
AN-FR-003	MUST	Maintain a versioned support matrix for DHIS2 2.41-2.44 expression functions, constructs, variable syntax and semantics.
AN-FR-004	MUST	Reject unknown expressions and missing metadata references with actionable diagnostics; never silently return zero.
AN-FR-005	MUST	Compile valid definitions into a typed, safe internal expression AST with explicit null, numeric, date and boolean semantics.
AN-FR-006	MUST	Track dependency graphs so changed events or metadata invalidate only affected indicators/ partitions.
AN-FR-007	SHOULD	Support local derived metrics as separately governed definitions with source, owner, rationale and version.
AN-FR-008	MUST	Expose indicator explain plans and the exact definition version used for each result.

4.2 Data ingestion, state and quality

ID	Level	Requirement
AN-FR-009	MUST	Consume committed event, tracked entity, enrollment, relationship and approved aggregate updates from a durable change stream.
AN-FR-010	MUST	Deduplicate by immutable change identifier and source-record version and process updates exactly once in observable effect.
AN-FR-011	MUST	Apply retraction and replacement of prior contributions when an event changes, is deleted, moves period, changes org unit, or changes eligibility.
AN-FR-012	MUST	Preserve soft-deleted, quarantined, conflicted and rejected source statuses without conflating them with valid records.
AN-FR-013	MUST	Define policy for completed, active, overdue, cancelled and pending event/enrollment states per metric.
AN-FR-014	MUST	Handle concurrent capture, late-arriving data, delayed peer changes and deterministic replay after restart.
AN-FR-015	MUST	Log data-quality exceptions, parse failures, missing codes, out-of-range values and orphaned references.

ID	Level	Requirement
AN-FR-016	SHOULD	Expose indicator eligibility/exclusion counts alongside reported values where privacy rules permit.

4.3 Tally, tracker and programme indicator computation

ID	Level	Requirement
AN-FR-017	MUST	Support event count, enrollment count, distinct tracked-entity count, episodes, admissions, diagnoses, referrals, outcomes and relevant stock movement tallies as separate grains.
AN-FR-018	MUST	Support counts, filtered counts, sums, minima, maxima, means, ratios, percentages, cohort retention and validated standard deviation/ variance where declared.
AN-FR-019	MUST	Define denominator-zero behaviour, missing-data treatment and rounding per metric, not by arbitrary frontend defaults.
AN-FR-020	MUST	Support repeatable stages, latest-per-enrollment selection, stage-specific event attribution and explicit temporal windows.
AN-FR-021	MUST	Prevent double-counting across service points when a clinical encounter involves multiple stages or devices.
AN-FR-022	MUST	Support standardised diagnosis grouping via governed local-to-ICD mapping and record unknown/ unmapped codes separately.
AN-FR-023	MUST	Calculate all-cause and cause-specific mortality from defined death events/outcomes and cause attribution, with case-fatality defined separately.
AN-FR-024	SHOULD	Support multi-stage episode-derived measures such as referral completion, lab turnaround and prescription-to-dispensing times.
AN-FR-025	MUST	Mark approximate, provisional or incomplete measures explicitly and disallow publication without a configured quality gate.

4.4 Periods, attribution and dimensions

ID	Level	Requirement
AN-FR-026	MUST	Provide canonical period generators for day, week, month, quarter, half-year and year with explicit time zone and calendar rules.

ID	Level	Requirement
AN-FR-027	MUST	Support configured DHIS2-compatible week variants and fiscal-year types only after version-specific verification.
AN-FR-028	MUST	Use half-open local instant windows [start, end) internally to avoid overlapping boundary records.
AN-FR-029	MUST	Support date-of-event, date-of-enrollment, incident date, discharge date and other approved attribution bases per metric.
AN-FR-030	MUST	Support org unit at event, enrollment, registration and authorised location rollups with explicit rules.
AN-FR-031	MUST	Provide sex, age-at-event, age-at-outcome, diagnosis, result, service point, clinic, stage, ward and programme dimensions where metadata permits.
AN-FR-032	MUST	Use governed category groupings; reject overlapping additive age bands and unknown/other omission.
AN-FR-033	SHOULD	Support comparable period-over-period and time-series queries including partial-period warnings.

4.5 Analytics -> aggregate reporting

ID	Level	Requirement
AN-FR-034	MUST	Store versioned mappings from indicator/measure to dataElement + categoryOptionCombo + attributeOptionCombo + orgUnit + period.
AN-FR-035	MUST	Generate proposals to prefill DER Aggregate Capture rather than directly marking datasets complete.
AN-FR-036	MUST	Maintain human review, approval, source provenance, override reason and locking for regulated submissions.
AN-FR-037	MUST	Keep observed tracker-derived values and user-entered aggregate reported values separate for reconciliation.
AN-FR-038	MUST	Never duplicate sums of overlapping clinic sections or disaggregations to produce hospital totals.
AN-FR-039	SHOULD	Support differences against central DHIS2 analytics, dataValueSets and approved hospital submissions.

ID	Level	Requirement
AN-FR-040	MUST	Publish only via a mapping compatible with the target DHIS2 version and stable metadata UIDs.

4.6 Query, drilldown and visual application

ID	Level	Requirement
AN-FR-041	MUST	Provide dashboard, chart, pivot, disaggregation table, patient-safe drilldown and record detail navigation as separate views.
AN-FR-042	MUST	Support navigation hierarchy: overview > care setting > programme > stage/section > measure > dimension cell > source record when authorised.
AN-FR-043	MUST	Include filters for periods, units, programme, stage, site, service point, category/diagnosis and quality status.
AN-FR-044	MUST	Support sorting, paginated result sets, search, saved views and copy/export under per-user access controls.
AN-FR-045	MUST	Explain cell numerator, denominator, count grain, period range, included states, exclusions and last updated timestamp.
AN-FR-046	MUST	Display aggregation freshness, metadata revision, source version, completeness, sync state and reconciliation state.
AN-FR-047	MUST	Use privacy suppression and protected export for small clinical counts and identifiers.
AN-FR-048	SHOULD	Offer charts (trend, bar, stacked bars, line, pivot, heatmap) selected by measure/dimension semantics.
AN-FR-049	SHOULD	Enable scheduled/triggered site reports through authorised operational scheduling when deployed.

4.7 Security, audit and operations

ID	Level	Requirement
AN-FR-050	MUST	Apply organisation-unit and programme authorisation to the underlying query and drilldown, not solely frontend filters.
AN-FR-051	MUST	Implement row/field-level protection, purpose-based audit and export logging for personally identifiable clinical data.

ID	Level	Requirement
AN-FR-052	MUST	Encrypt data in transit and at rest using the platform security architecture and protect analytics caches/backups.
AN-FR-053	MUST	Provide resilient queue, retry, dead-letter quarantine, replay, metrics and alerting for failed computations.
AN-FR-054	MUST	Allow recomputation following changed metadata, corrections, time-zone configuration or backfilled records.
AN-FR-055	MUST	Support tenancy isolation per site with explicit cross-site visibility only after central authorisation.
AN-FR-056	SHOULD	Offer incremental recomputation progress, worker capacity limits and rate-limited complex queries.
AN-FR-057	MUST	Preserve user-visible decision provenance for every aggregate cell proposed for submission.

5. Period type catalogue and date semantics

Date representation: Use `date` for clinical event dates and `timestamp with time zone` for writes/ingestion. Convert event date into facility-local period boundaries, then use `[startInstant, endInstant)` comparisons. Never assume UTC midnight corresponds to local midnight. A metric may explicitly use event date, enrollment date, incident date or discharge date. Record that basis with the result.

Logical period	Example	Intended behaviour	Compatibility handling
Daily	`20261110`	Local calendar day	Normalise DHIS2 period syntax at adapter
Weekly ISO	`2026W46`	ISO Monday-Sunday	Explicitly name ISO week system
Weekly variants	Saturday/Sunday/ Wednesday/Thursday start	Where mandated	Verify exact target DHIS2 period identifiers before enabling
Biweekly	Two-week period	Only when site reporting requires	Align with native target semantics
Monthly	`202611`	Calendar month	Core HMIS mapping
Bimonthly	`202611B` (illustrative only)	Two-month window	Never send illustrative ID without validation
Quarterly	`2026Q4`	Three calendar months	Core
Six-monthly	`2026S2`	Six calendar months	Check native semantics
Yearly	`2026`	Calendar year	Core
Financial years	Site-specific start month	Fiscal boundary window	Store explicit start/end and target-code mapping

Logical period	Example	Intended behaviour	Compatibility handling
Rolling windows	Last 7/30/90 days	Dynamic dashboards	Derived local window, not automatically a DHIS2 period
Custom range	2026-10-01 to 2026-11-10	Ad hoc analysis	Not directly an aggregate dataset period unless mapped

Period examples: A consultation recorded on 2026-11-10 local time belongs to November 2026, Q4 2026 and ISO week 46. A correction to an event date in a different month must retract the prior month contribution and insert into the new month. District or organisational rollups must not re-add the same patient's activity twice when it was already aggregated at a parent level. Some DHIS2 period syntax differs by release or period variant; adapter unit tests must verify every enabled format.

5.1 Supported dimension defaults

- **Care setting:** OPD, IPD, Emergency, Maternity, diagnostics, other approved departments.
- **Clinical programme:** the actual DHIS2 program UID (not display name) and stage UID.
- **Person dimensions:** sex/gender according to configured meaning, age at relevant anchor date, documented residence, clinical population grouping when lawful.
- **Clinical conditions:** diagnosis code, case status, underlying cause, secondary cause, clinical outcome, case type (new/revisit), severity.
- **Operational dimensions:** service point, source event org unit, enrolment org unit, ward, team, service type, facility ownership if configured.
- **Reporting dimensions:** period, organisation unit, indicator and categoryOptionCombo where mapped.
- **Quality dimensions:** pending, verified, conflicted, corrected, missing, approved, reconciled.

Cross-cutting age-group definitions must not be conflated with unknown values; `Unknown` remains a visible category and excluded from age-stratified denominators only when explicitly specified.

6. Calculation semantics and correctness contract

6.1 Metric definition schema

Every derived metric must specify: `id`, `name`, `code`, `owner`, `grain`, `programUid`, `stageUids`, `aggregationType`, `analyticsType`, `expressionAst`, `filterAst`, `dateAnchor`, `orgUnitAnchor`, `sourceStates`, `distinctKey`, `dimensions`, `denominator`, `unit`, `nullPolicy`, `roundingPolicy`, `minCellSize`, `metadataRevision`, `effectivePeriod`, `mappingStatus`, `sourceLineagePolicy` and `version`.

A metric is **not aggregatable** unless `grain` and `aggregation` are compatible: a mean of site-level averages must be weighted by their source counts, and a distinct-patient count cannot be summed across arbitrary sites to obtain national unique patients.

6.2 Event vs enrollment indicator behaviour

DHIS2 distinguishes event-type program indicators from enrollment-type indicators. Event analytics is typically anchored to event org unit and event date; enrollment analytics is typically anchored to enrollment org unit and enrollment date. Enrollment analytics may reference events across stages, and a repeatable stage can have special latest-event semantics. DER must use the definition and target version's semantics, not a generic count of event rows. Any ambiguity triggers an unsupported/verification state rather than an apparently successful result.

Conceptual execution order: determine eligible org unit/period, filter candidate records, evaluate expression at the configured grain, then apply aggregation; for downstream ratio indicators evaluate numerator and denominator according to their separate definitions. Custom aggregation functions require verified semantics.

6.3 Clinical counting contracts

Metric	Correct source grain	Date basis	Primary deduplication/ interpretation
OPD attendances	Completed OPD encounter	Encounter date	Unique encounter UID, not every clinical stage
Unique OPD patients	Patient UID	Encounter date	Count distinct within period/site, not sum of weekly distinct counts
New OPD disease cases	Valid qualifying encounter/ case definition	Case or encounter date	New-case rule and diagnosis role required
IPD admissions	Admission episode	Admission date	One per admission, not per ward transfer

Metric	Correct source grain	Date basis	Primary deduplication/ interpretation
IPD discharges	Admission episode	Discharge date	Deaths, referrals and absconded outcomes distinctly classified
Inpatient deaths	Deceased admission episode	Death date	Explicit death outcome and cause attribution
Diagnoses	Primary diagnosis per defined encounter/episode	Encounter/discharge date	Do not sum all listed secondary diagnoses as unique cases
Lab tests performed	Test-level service unit	Result/perform date	Request with multiple tests expands to distinct tests
Medication dispensed	Authoritative dispensing line transaction	Actual dispense date	Replays never deduct or tally twice
Maternal deliveries	Delivery episode	Delivery date	Multiple births count one delivery but multiple infants
Live births	Individual live-born infant	Birth date	Twins count as two live births
Stillbirths	Birth outcome event	Birth date	Apply governed gestational-age/weight threshold
Perinatal mortality	Stillbirth/early neonatal criteria	Birth/death dates	Follow approved national definition
Referrals completed	Completed referral request	Completion date	Do not count task attempts or duplicate handoffs

6.4 Example correctness vectors

Vector A - changed event: Event E1 initially counts for October. Its date is corrected to November. October count becomes 0 and November count becomes 1. Replaying the same correction again changes nothing.

Vector B - visit with 4 stages: A single OPD encounter has registration, triage, consultation and lab stage events. OPD encounter tally = 1, stage events = 4, laboratory test tally depends on test lines. Distinct patients = 1.

Vector C - two visits: Patient P1 attends twice in November. OPD attendances = 2; distinct patients = 1. Summing daily distinct counts could yield 2 and is prohibited for monthly unique-patient statistics.

Vector D - mortality: One inpatient admission ends in death with documented underlying cause and two secondary diagnoses. Inpatient deaths = 1; underlying-cause row increments once; secondary diagnoses are not treated as additional deaths.

Vector E - denominator zero: In a ward with zero discharges, inpatient mortality percentage for discharge episodes is **undefined** (shown as “No denominator”), not 0%.

Vector F - provisional conflict: An event is disputed between two peers. Keep last reconciled committed contribution and label potential freshness/quality impact according to the chosen conflict policy; do not count both versions.

7. Engine architecture and processing flow

7.1 Recommended server modules

1. **Metadata Synchroniser:** Snapshot owner, UID, options, type, programme/stage and indicator definitions. Detect semantic diff and affected lineage.
2. **Expression Lexer/Parser:** Parse only allowed DHIS2 syntax for each version; generate AST; type-check and annotate dependencies.
3. **Expression Executor:** Evaluate typed expression per event or enrollment snapshot with deterministic null and date handling.
4. **Incremental Change Consumer:** Read durable change journal, claim update atomically, restore previous contribution and apply new one.
5. **Contribution Ledger:** Store one stable, versioned contribution per metric, source/grain key, period and dimension cell.
6. **Aggregation Materializer:** Maintain count/sum/mean-state/min/max or rebuild where retractions are nontrivial; distinct counts need explicit exact-set support or clear approximation labels.
7. **Period Resolver:** Produce canonical time windows and target-compatible DHIS2 period identifiers.
8. **Dimension Resolver:** Map age groups, diagnosis values, clinical statuses, category combinations and service ownership.
9. **Quality and Reconciliation:** Compute source-completeness, mismatched aggregates, failed expressions, late data and audit status.
10. **Query Planner and API:** Apply authorisation first, aggregate or paginate, and return data with source lineage and freshness.
11. **Aggregate Proposal Mapper:** Create DER Aggregate Capture prefill proposals with review state, not direct writes.
12. **Observability and Control:** Lag, throughput, queue depth, stale partitions, errors, replay controls, snapshot diagnostics.

7.2 State model

```
stateDiagram-v2
 [*] --> Imported
 Imported --> Parsed
 Parsed --> Unsupported: Unknown function/schema
 Parsed --> Ready: Valid definition
 Ready --> Computing
 Computing --> Provisional
 Provisional --> Verified: Quality gate
 Verified --> Proposed: Eligible aggregate mapping
 Proposed --> Approved: HMIS review
 Approved --> Published: DHIS2 accepted
 Published --> Reconciled: Same expected values
 Published --> Diverged: Values differ
 Diverged --> Computing: Investigate/recompute
 Provisional --> Computing: Source edit or metadata change
```

7.3 Database projection schema (illustrative, DER-owned)

```
CREATE TABLE analytics_definition (
  id UUID PRIMARY KEY, source_uid VARCHAR(24), code VARCHAR(32),
  definition_version INTEGER NOT NULL, metadata_hash TEXT NOT NULL,
  analytics_type TEXT NOT NULL, aggregation_type TEXT NOT NULL,
  compiled_ast JSONB, support_state TEXT NOT NULL,
  definition JSONB NOT NULL, effective_from TIMESTAMPTZ NOT NULL
);
CREATE TABLE analytics_source_cursor (
  site_id UUID NOT NULL, stream TEXT NOT NULL, cursor BIGINT NOT NULL,
  PRIMARY KEY (site_id, stream)
);
CREATE TABLE analytics_contribution (
  site_id UUID NOT NULL, metric_id UUID NOT NULL,
  definition_version INTEGER NOT NULL, source_key TEXT NOT NULL,
  source_version TEXT NOT NULL, period_key TEXT NOT NULL,
  org_unit_uid TEXT NOT NULL, dimension_key TEXT NOT NULL,
  contribution_value NUMERIC, eligibility BOOLEAN NOT NULL,
  source_state TEXT NOT NULL, updated_at TIMESTAMPTZ NOT NULL,
  PRIMARY KEY(site_id, metric_id, definition_version, source_key, period_key, org_unit_uid,
        dimension_key)
);
CREATE TABLE analytics_cell (
  site_id UUID NOT NULL, metric_id UUID NOT NULL,
  definition_version INTEGER NOT NULL, period_key TEXT NOT NULL,
  org_unit_uid TEXT NOT NULL, dimension_key TEXT NOT NULL,
  count_value BIGINT, sum_value NUMERIC, value NUMERIC,
  freshness_at TIMESTAMPTZ, quality_state TEXT NOT NULL,
  PRIMARY KEY(site_id, metric_id, definition_version, period_key, org_unit_uid, dimension_key)
);
CREATE TABLE analytics_publication (
  proposal_id UUID PRIMARY KEY, metric_id UUID NOT NULL,
  dataset_uid TEXT NOT NULL, data_element_uid TEXT NOT NULL,
  category_option_combo_uid TEXT NOT NULL, attribute_option_combo_uid TEXT NOT NULL,
  period_key TEXT NOT NULL, org_unit_uid TEXT NOT NULL,
  proposed_value NUMERIC, reviewed_value NUMERIC,
  state TEXT NOT NULL, provenance JSONB NOT NULL
);
```

Implementation note: The simplified contribution key must be extended with a stable contribution-discriminator for definitions that emit multiple cells per source record (e.g., multivalued clinical dimensions). The ingestion transaction must atomically update contribution history and enqueue affected aggregate cells. Non-invertible aggregation states (exact distinct, min/max) may need selective partition recomputation rather than a naive subtraction. Apply retention, indices, access controls, partitioning and encryption in the actual database migration.

7.4 Incremental algorithm

```
onCommittedSourceChange(change):
  verifySignatureAndSiteOwnership(change)
  if journalVersionAlreadyApplied(change.id, change.version): return NO_OP
  affectedMetrics = dependencyGraph.forSource(change.sourceType, change.changedFields)
  transaction:
    for metric in affectedMetrics:
      prior = contributionLedger.find(metric, change.sourceUid)
      removeOrInvalidate(prior)
      current = loadCommittedVersion(change.sourceUid, change.version)
      if eligible(metric, current):
        contributions = execute(metric, current)  // zero-to-many cells
        upsertContributions(contributions)
      enqueueRebuildOrDelta(prior, contributions)
    advanceCursorAndCommit(change)
  asynchronouslyRecomputeQueuedCells()
  publishFreshnessAndQualityStatus()
```

Critical safety rules: do not materialise across uncommitted source transactions; never acknowledge journal cursor before durable processing; use bounded retries and dead-letter queues; store job reasons and version hashes; consider patient merge/identity correction as full dependent partition invalidation.

8. Analytics-to-aggregate mapping

DER shall read the 15 parent dataset definitions, their 175 native sections and DER logical-subsection manifest. The *mapping* is a governed object independent of DHIS2's metadata JSON. A valid mapping must specify the computation grain and the exact target cell, including category option combo and attribute option combo; period type must match the parent dataset. One target cell must have one authoritative publication route.

Example conceptual mapping (identifiers shown as placeholders, NOT importable DHIS2 UIDs):

```
{
  "mappingId":"example-opd-attendance",
  "sourceMetricCode":"OPPI0001",
  "sourceGrain":"ENCOUNTER",
  "periodType":"Monthly",
  "dateAnchor":"encounterDate",
  "orgUnitAnchor":"eventOrgUnit",
  "target":{
    "datasetUid":"<actual-dataset-uid>",
    "dataElementUid":"<actual-data-element-uid>",
    "categoryOptionComboUid":"<resolved-coc-uid>",
    "attributeOptionComboUid":"<resolved-aoc-uid>"
  },
  "publicationMode":"REVIEW_REQUIRED",
  "overlapPolicy":"EXCLUSIVE_CELL_OWNER"
}
```

Review behaviour: Compare system-derived numerator with manual section totals; show coverage and potential double counting; flag a partial month or pending sync; capture review reason and approver; update only the intended DER aggregate workspace cell. Keep reported values frozen after approval unless formally reopened and audited. An approved aggregate workspace still uses native DHIS2 `dataValueSets` submission rules.

9. API contract (DER-owned, illustrative)

All endpoints are namespaced under `/api/analytics/v1` and follow normal session/token authorisation. The examples describe the proposed DER API, **not existing DHIS2 API endpoints**.

Method	Path	Purpose
GET	`/capabilities`	Supported functions, periods, dimensions and version compatibility
GET	`/metrics`	Browse authorised metrics, definitions and support states
GET	`/periods?type=Monthly&year=2026`	Enumerate canonical period windows
POST	`/query`	Analytic matrix with filters, group-by, pagination and freshness
POST	`/timeseries`	Trend queries and partial-period flags
POST	`/drilldown`	Authorised contributing source rows and exclusions
GET	`/explain/{metricId}`	Formula, filter, grain, definition revision and support status
GET	`/quality`	Completeness, data drift, lag and error summary
POST	`/exports`	Audited CSV/XLSX/PDF output job
POST	`/aggregate-proposals`	Propose cells for DER Aggregate Capture review

Method	Path	Purpose
GET	`/reconciliation`	Compare local, aggregate and central snapshots
POST	`/admin/recompute`	Protected recalculation by scope and reason
GET	`/admin/jobs`	Rebuild progress, queue lag, errors

Example query:

```
{
  "metrics":["OPPI0001"],
  "period":{"type":"Monthly","id":"202611"},
  "orgUnits":["<authorised-orgunit-uid>"],
  "dimensions":["sex","ageGroup","servicePoint"],
  "filters":{"program":"<program-uid>","sourceStatus":"COMMITTED"},
  "page":{"limit":100,"cursor":null}
}
```

A response includes `metricUid`, `period`, `periodWindow`, `definitionVersion`, `sourceSnapshotCursor`, `qualityState`, `freshnessAt`, `units`, cell values and `drilldownToken`. The drilldown token is an opaque short-lived server-side scope identifier; clients cannot forge filters to request broader access. Query input must be constrained to authorised dimensions and valid expressions.

10. Modern DER Analytics user experience

10.1 Application placement and navigation

DER Analytics launches from the DER app launcher in its own full-screen application shell. It uses the same visual design system as DER Capture: calm flat surfaces, restrained dividers, legible dense tables, no heavy borders/ gradients, consistent spacing, keyboard support, accessible colours and predictable navigation. **Do not mix source lists and detailed forms on the same page.**

Navigation:

1. **Overview:** site analytics, period context, data freshness, configurable KPIs, alerts, pending validations.
2. **Explore:** dataset/programme selector, reusable dimensions, group-by, time-series, pivot, chart builder.
3. **Program Indicators:** searchable catalogue, supported/unsupported chips, definition inspection and execution trace.
4. **Tracker & Event Tally:** new events, visits, enrollments, stage throughput and case counts, segmented by time and service points.

5. **Hospital Reports:** the 15 parent datasets, native section and DER subsection coverage, calculated contributions and readiness.
6. **Drilldowns:** authorised list of contributing visits/enrollments/events, then a dedicated record detail page.
7. **Saved Analytics:** user/team saved views, shared dashboards, exports and comparison snapshots.
8. **Quality & Reconciliation:** incomplete records, mismatches, unmet formula dependencies and central sync status.
9. **Administration:** capability matrix, metadata versions, query limits, rebuild jobs, mapping configuration, access auditing.

10.2 Page architecture and interactions

Overview page - title, site/period badge, freshness timestamp, 4-6 configurable KPI cards, one main trend chart, a facility summary table and quality issues. A KPI click opens the metric detail, not an arbitrary chart. Empty/loading/error/no-permission states are distinct.

Metric detail page - metric name, definition, value, period, data-as-of, time series, stratified breakdown, confidence/quality warnings; clicking a non-suppressed cell opens drilldown. The calculation explanation is visible without leaving the page.

Explore builder page - two-panel mode: compact query controls (metric, time, program, org unit, dimensions, status) and a sizeable results area. Chart/pivot switches preserve filters. For results over thousands of cells, server-side pagination/virtualization is mandatory.

Hospital reporting page - dataset > service group > native section > logical subsection. Show source count, computed count, reported value, difference, completion and reviewer. Device-group assignment is visible as context, not an analytics permission bypass.

Drilldown page - header with exact originating chart cell and filter breadcrumbs; source rows with event date, clinical service, masked identity when appropriate, data state, provenance and navigation to record. When privacy threshold is not met, show a suppression explanation instead of identities.

Record detail page - dedicated view through authorised DER Capture route, never embedding editable capture form in the analytics table. Maintain return navigation to the originating metric and filter context.

UX interactions: deep links, saved filters, hover descriptions, patient-safe field masking, export confirmation for potentially sensitive data, keyboard and screen reader support, responsive tablets, no reliance on colour alone, theme consistency, print-friendly reporting.

10.3 Navigation state

Route format (illustrative): `/analytics/overview`, `/analytics/explore`, `/analytics/metrics/:metricId`, `/analytics/reports/:datasetUid`, `/analytics/drilldown/:scopeToken`, `/analytics/quality`, `/analytics/admin`. URL state holds non-sensitive filters; private record IDs and PHI must not be exposed in shareable query strings.

11. Drilldown and privacy contract

Drilldown is **not** a generic reverse SQL query from an aggregate count. It is a lineage-based, authorised operation with an explicit relationship to the metric version and the exact cell filters. Only appropriate metrics have record-level drilldowns; aggregate-only data submitted by a separate clinic might support a source document/reference but no patient row.

- All authorisation is rechecked by the server on every drilldown page and export request.
- Aggregate viewers can see numeric summaries, but only users with patient-record privileges and a permitted clinical purpose can open patient detail.
- Minimal necessary fields only; pseudonymisation/masking by default in management reports.
- Small-cell suppression threshold is configurable; suppressed cells must not be inferred through complementary totals, exports or repeated dimension queries.
- Count uniqueness guarantees apply only to the scoped site and definition; remote deduplication remains unresolved while offline.
- Audit filter changes, sensitive reads, exports, records opened and impersonation/break-glass actions.
- Never cache PHI in shared browser storage or service worker caches without specifically approved encryption and expiry controls.

12. Reconciliation with central DHIS2

DHIS2 central APIs can provide event analytics, tracked-entity analytics, program indicator values and aggregate analytics; central analytics tables may be refreshed asynchronously and may not include locally pending data. DER Analytics must compare only like-for-like metric definitions, periods, organisation units, statuses and snapshots; mismatches have classifications such as `LOCAL_PENDING_SYNC`, `CENTRAL_ANALYTICS_NOT_REFRESHED`, `METADATA_VERSION_DRIFT`, `INCOMPATIBLE_FORMULA`, `ORGUNIT_ATTRIBUTION`, `DATE_ANCHOR`, `DEDUP_POLICY`, `LATE_DATA`, `REJECTED_TRACKER_UPDATE` and `REAL_MISMATCH`.

A parity test is valid only when the upstream DHIS2 instance has acknowledged the relevant source revisions, the analytics refresh is complete, and the compared indicator and dimension semantics match. DER must not represent a local partial count as a verified replica of national totals.

13. Non-functional requirements and measurable targets

The following are *design acceptance targets*, subject to site hardware and realistic benchmark confirmation.

ID	Quality	Target / test method
AN-NFR-001	Availability	Reads for already-materialised metrics continue through WAN outage and central DHIS2 unavailability
AN-NFR-002	Latency	95% of ordinary single-site dashboard requests under 2 s on reference workload, after warm-up
AN-NFR-003	Freshness	95% of committed ordinary event updates reflected in relevant tally within 10 s on reference workload
AN-NFR-004	Replay correctness	At-least-once event delivery must produce exactly-once observable tally effect
AN-NFR-005	Durability	Loss of analytics worker must not lose journal events or approved publication audit records
AN-NFR-006	Calculation integrity	Golden fixture results exact for integer counts, approved decimal tolerance for rate/mean
AN-NFR-007	Privacy	No unauthorised record-level drilldown or export under privilege/ tenant tests
AN-NFR-008	Scale	Benchmark at 1 million events / 100,000 enrollments / 200 indicators without unbounded memory growth

ID	Quality	Target / test method
AN-NFR-009	Recovery	Worker restart and deterministic replay from committed cursor without count inflation
AN-NFR-010	Observability	Expose ingestion lag, query p95, failed definitions, dead letters and stale partitions
AN-NFR-011	Compatibility	Explicit tested matrix per DHIS2 2.41-2.44 expression/period/API adapter
AN-NFR-012	Accessibility	Keyboard navigable, WCAG 2.2 AA design target for supported screens
AN-NFR-013	Maintainability	Isolated parser, engine, storage and UI components with contract tests
AN-NFR-014	Security	Tenant/site isolation; encrypted transport and storage; least-privilege administration
AN-NFR-015	Data quality	Never publish unsupported, partial, conflicted or unmapped metrics as verified

14. Testing matrix and acceptance criteria

Test	Scenario	Expected result
AT-01	1 visit, 4 stage events	OPD attendance 1; stage counts 4; no duplicate person
AT-02	2 encounters, same patient	Attentions 2, distinct patient 1
AT-03	Event date changed across months	Old period retracted; new period inserted
AT-04	Event deletion/cancellation	Contribution removed under policy
AT-05	Duplicate journal delivery	No change to result
AT-06	Concurrent device writes	One canonical committed version contributes
AT-07	WAN offline then recovery	Local tally maintained; central reconciliation correctly labelled
AT-08	Enrollment PI across repeat stages	Correct semantics per validated target DHIS2 version
AT-09	Missing option code	Report error/unknown, never silently classify as other disease
AT-10	Unknown expression function	Definition unsupported with diagnostic, no fake 0
AT-11	Time-zone month boundary	Correct Kampala reporting month
AT-12	Patient identity merge	Distinct-patient count recomputed without duplication
AT-13	Death with primary and secondary causes	One death under assigned underlying cause
AT-14	Small cell drilldown	Suppression and server-side permission enforcement
AT-15	Export attempt without permission	Denied, audited
AT-16	Metadata change to indicator filter	Impacted cells invalidated, recomputed with revision

Test	Scenario	Expected result
AT-17	Aggregate proposal for unmapped cell	Rejected, not silently submitted
AT-18	Different aggregate and tracker values	Difference shown with status and provenance
AT-19	Repeated weekly-to-monthly rollup	Uniques recomputed, never summed blindly
AT-20	Materialisation worker crash during transaction	Recovery yields exact expected result

14.1 Reference workload and evaluation

Use synthetic (not real patient) fixtures spanning outpatient, inpatient, emergency, laboratory, maternal and specialised clinics, plus status transitions, updates, corrections, backfills, multi-device concurrency, outages and metadata drift. Record event count, stage distributions, per-indicator execution time, memory/CPU, queue depth, incremental throughput, dashboard query p50/p95/p99, correctness versus a simple reference batch implementation, and controlled parity against central DHIS2 after its analytics refresh.

15. Implementation phases and deliverables

Phase	Engine scope	App/QA deliverable	Exit gate
P1	Metadata snapshots, source journal consumer, period service	Catalog page and readiness diagnostics	Versioned read models and cursor replay tests
P2	Event count, enrollment count, stable-tally states	Overview, event/explore pages	Correct mutable-event retraction and drilldown
P3	Program indicator parser/AST and safe evaluator	Metric definition/unsupported details	Golden tests for verified DHIS2 functions
P4	Dimension rollups, exact distinct, cohort/temporal metrics	Pivot, charts, time series	Correct age/sex, date and site attribution
P5	Aggregate cell mapping, provenance, approval integration	Hospital reporting and reconciliation	No unapproved dataValueSet publication
P6	Security hardening, observability, performance and compatibility	Saved views, audit, admin controls	Load/outage/authorization and per-version sign-off

15.1 Software deliverables

- Versioned SRS with functional and non-functional requirements, traceability and threat model.
- Technical ADRs for local computation authority, expression language, rollup/grain and publication policy.
- Metadata compatibility manifest for DHIS2 2.41-2.44 and explicit unsupported features.

- Typed indicator parser/AST, evaluator, period library and reference test vectors.
- Durable ingestion, canonical projection, contribution ledger, aggregation materializer and reconciliation services.
- Authenticated REST API with OpenAPI specification, limits and error codes.
- Full DER Analytics web UI, dashboard, Explore, metric detail, drilldown, reporting, quality and admin views.
- Load/fault/permission/parity tests, synthetic fixtures, deployment charts/configurations and runbooks.

16. DHIS2 interoperability references

Authoritative references to verify during implementation:

1. DHIS2, *Tracker Analytics - System Design*, https://docs.dhis2.org/en/implement/database-design/tracker-system-design/tracker-analytics.html - event and enrollment indicator granularity, organisation unit/period interpretation and repeatable stages.
2. DHIS2, *Configuring Programs (2.43)*, https://docs.dhis2.org/en/use/user-guides/dhis-core-version-243/configuring-the-system/programs.html - program indicator components, filters and aggregation types.
3. DHIS2, *Analytics Web API (2.42)*, https://docs.dhis2.org/en/develop/using-the-api/dhis-core-version-242/analytics.html - analytics, event query, aggregate dimensions and API shape.
4. DHIS2, *Analytics Web API (2.41)*, https://docs.dhis2.org/en/develop/using-the-api/dhis-core-version-241/analytics.html - earlier-version reference for compatibility testing.
5. DER Hospital Consolidated Aggregate Reporting metadata, locally supplied in this conversation, containing 15 parent datasets and 175 sections (baseline only; full implementation requires actual import validation).

Clinical standards note: WHO routine facility reporting and ICD-11, LOINC, SNOMED CT, ATC, UCUM and approved national HMIS classifications may guide the indicator and terminology governance layer, but DER should not assert a standard terminology match without an approved and versioned mapping.

17. Final acceptance decision

Release DER Analytics to site production only after the following are all demonstrated: source edits reliably retract prior contributions; indicator semantics are explicitly supported for the target DHIS2 version; patient-level drilldown access is enforced on the server; aggregated proposals cannot bypass local review; central-versus-local differences can be explained; periods and age groups are correct; and an outage/replay test produces identical results to the verified baseline.